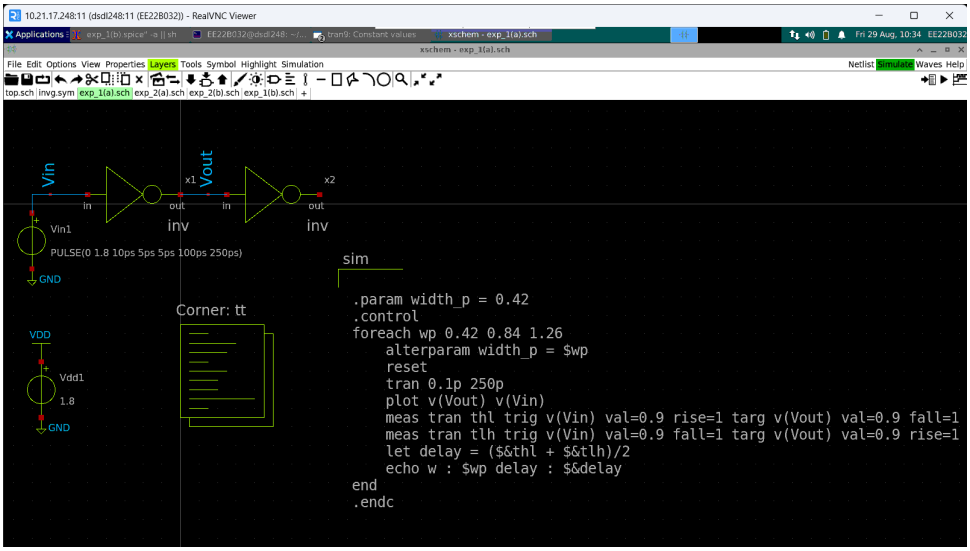
EE5311: Digital IC Design
Tutorial 4 Report

Putta Shravya
Roll Number: EE22B032

August 31, 2025

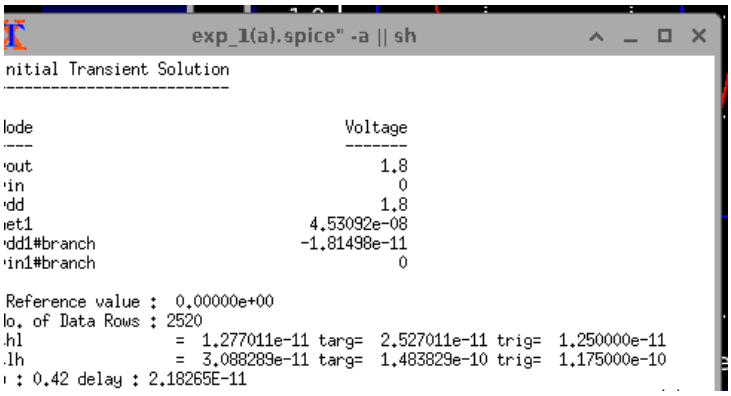
Question 1

Part A

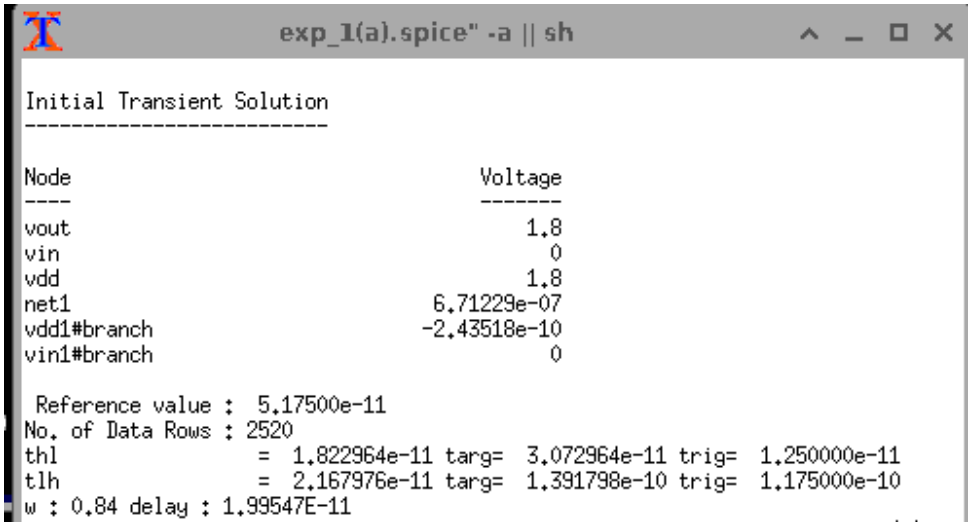


Schematic

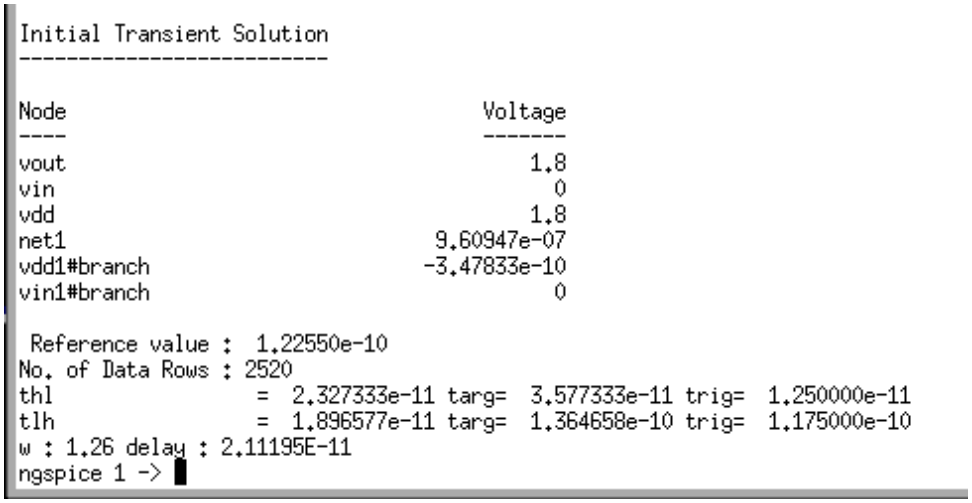
Delay



width=0.42

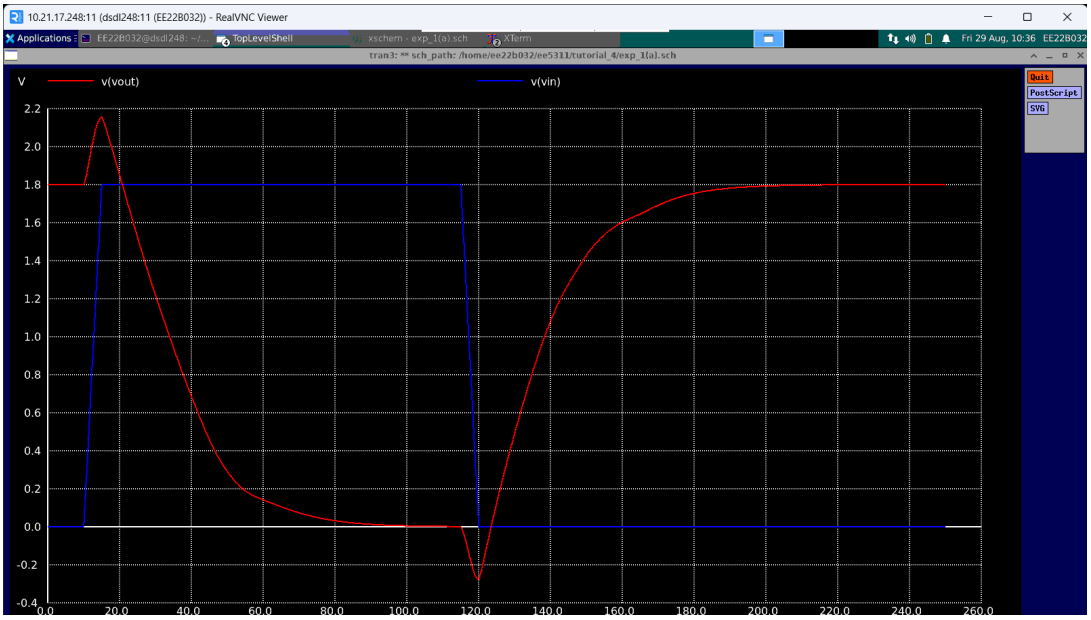


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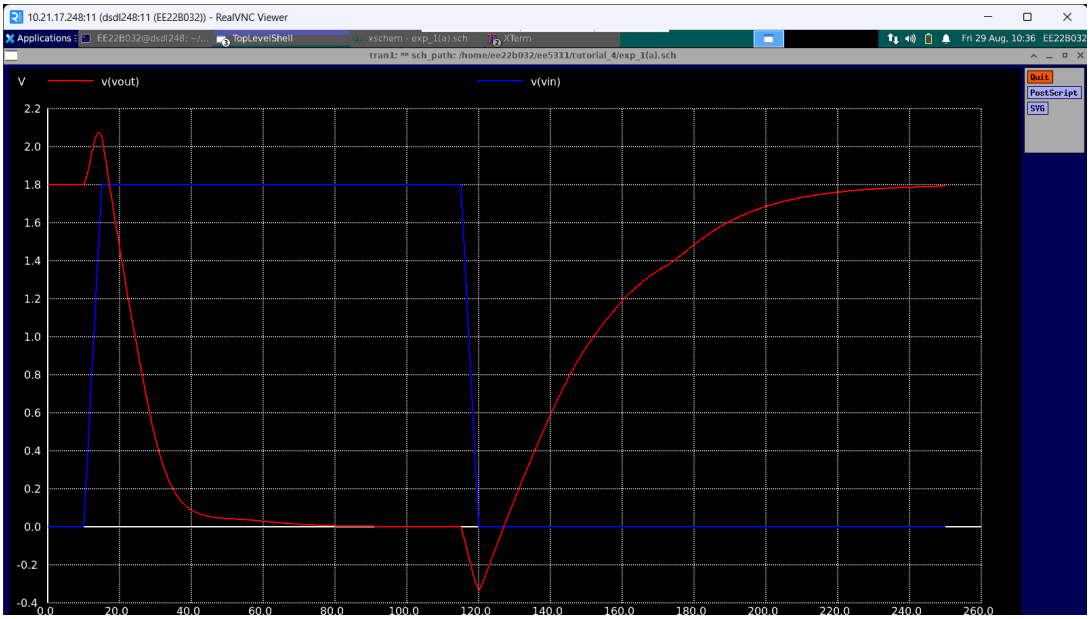


width=1.26

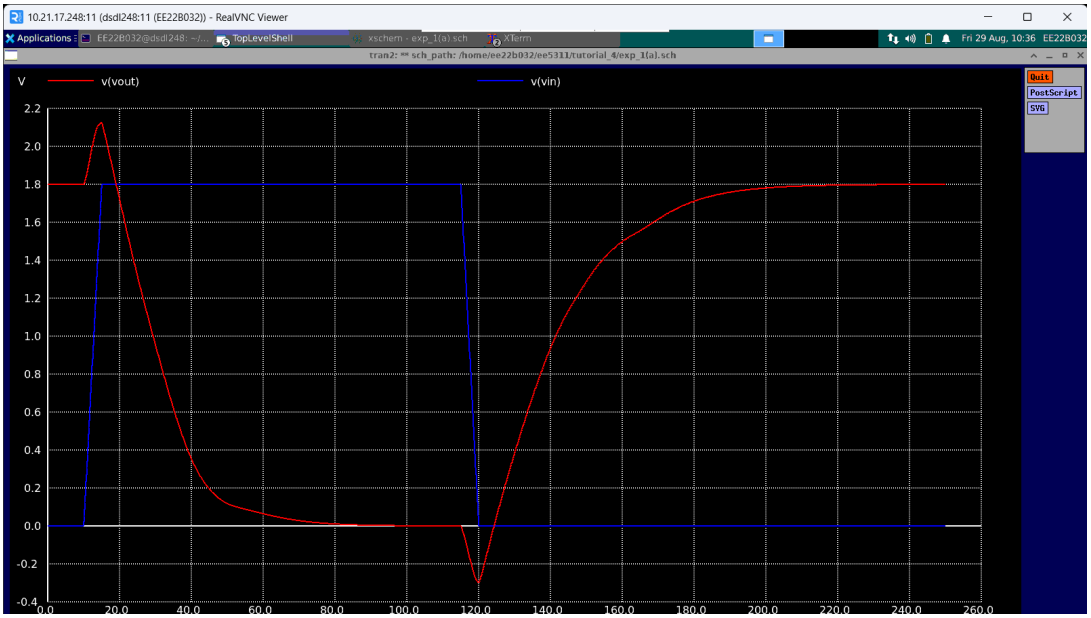
Transient Characteristics



width=0.42

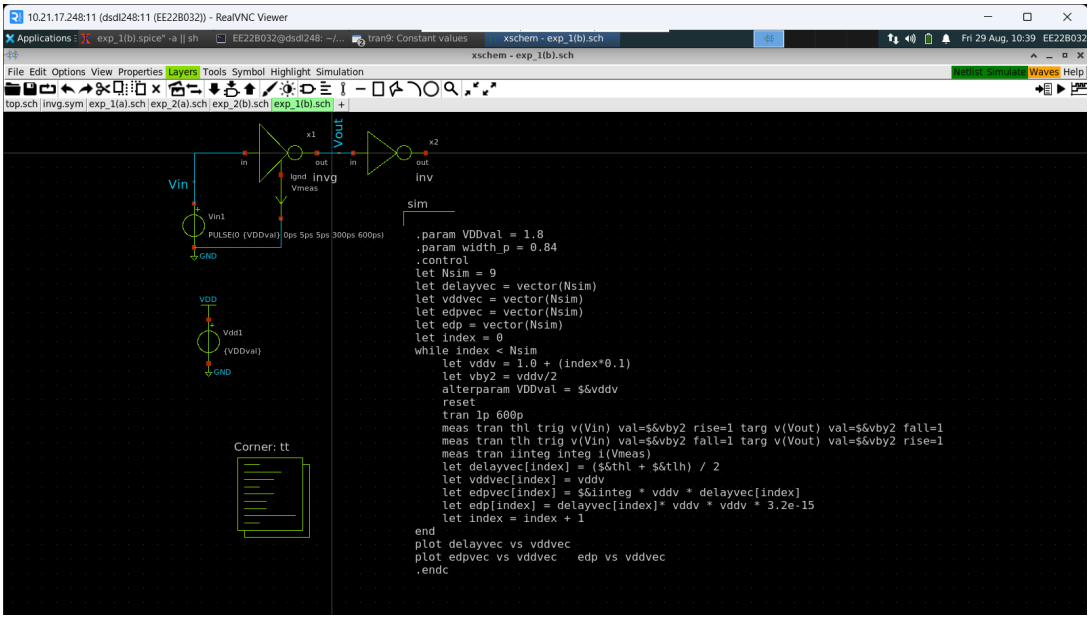


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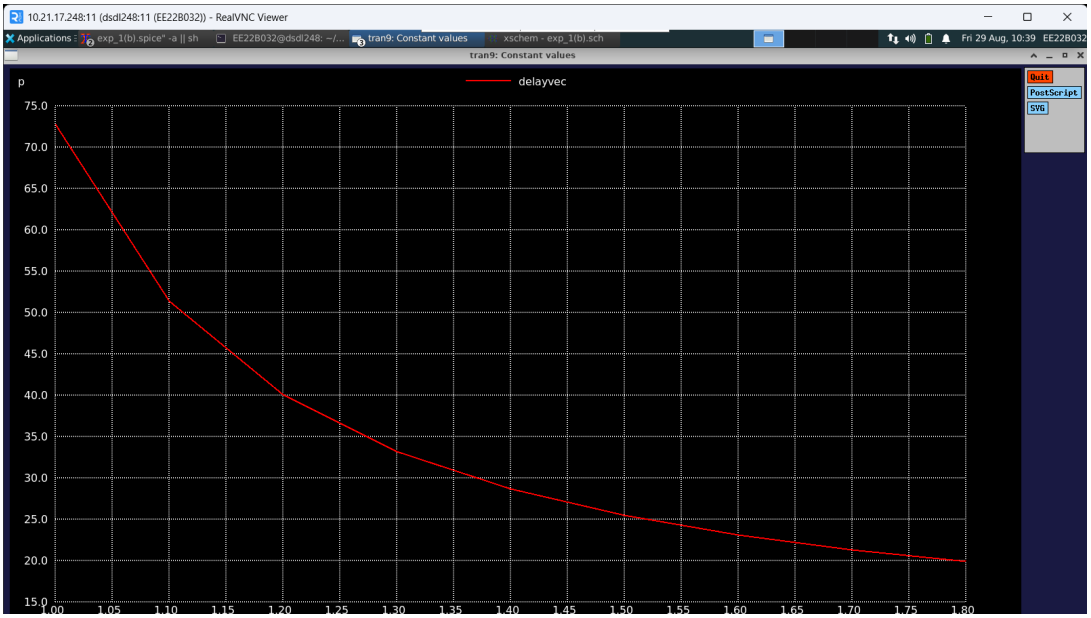


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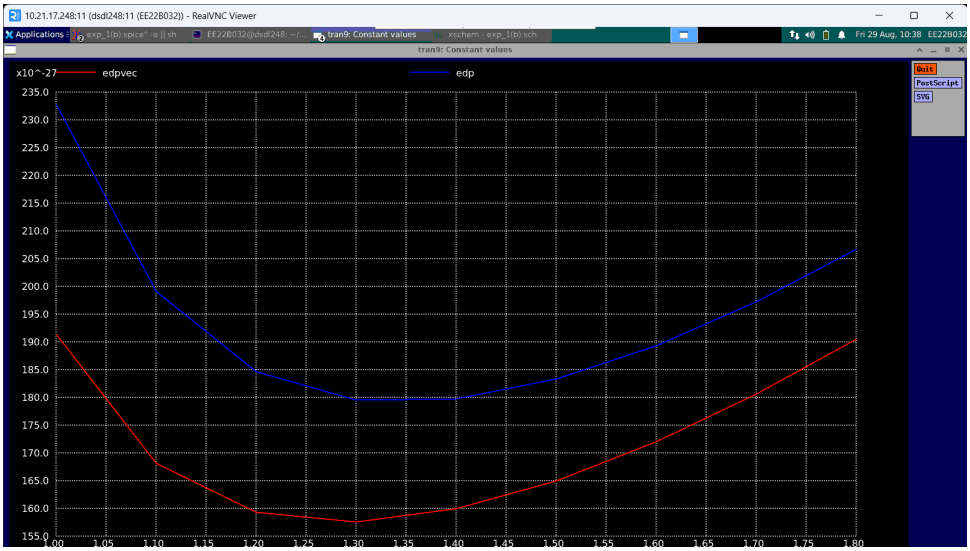
Part B,C



Schematic



Delay Plot



Energy Delay Product Plot

1b) from part a we have observed that for $W_p = 0.84\mu m$ the delay is minimised ($W_n = 0.42\mu m$).

t_{PLH} : PMOS is in saturation
NMOS is in cutoff

$$= \frac{C_{total} V_{DD}/2}{I_{Dsat}}$$

t_{PHL} : PMOS is in cutoff
NMOS is in saturation

$$= \frac{C_{total} V_{DD}/2}{I_{Dsat}}$$

$$I_{Dsat_p} = \frac{\mu_p C_{ox}}{2}\left(\frac{W}{L}\right)_p \frac{E_{cp}L_p (V_{DD} - |V_{Tp}|)^2}{E_{cp}L_p + V_{DD} - |V_{Tp}|}$$

$$I_{Dsat_n} = \frac{\mu_n C_{ox}}{2}\left(\frac{W}{L}\right)_n \frac{E_{cn}L_n (V_{DD} - V_{Tn})^2}{E_{cn}L_n + V_{DD} - V_{Tn}}$$

$C_{total} = 3.2fF$

$E_{cp}L_p = 1V$

$E_{cn}L_n = 0.96V$

$$t_{PLH} = \frac{3.2\times10^{-15}}{2} \times \frac{V_{DD}}{205.63\times10^{-6}} \frac{(V_{DD}+0.3)}{(V_{DD}-0.7)^2}$$

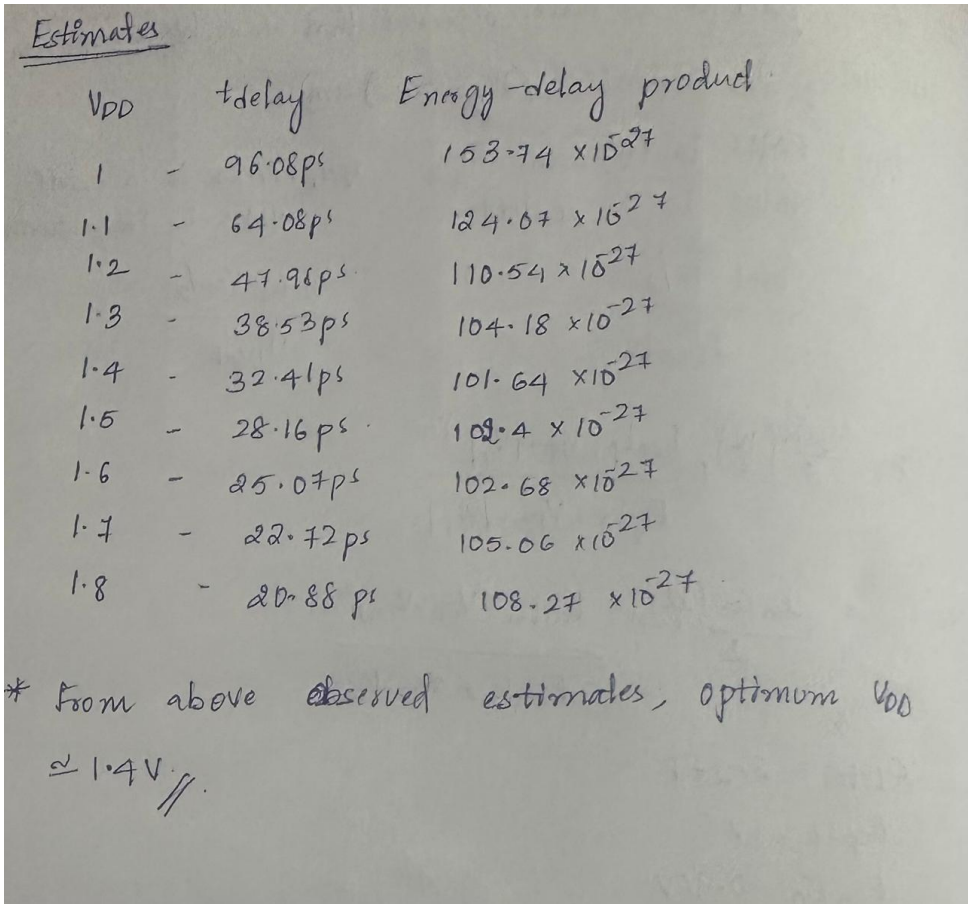
$$t_{PHL} = \frac{3.2\times10^{-15}}{2} \times \frac{V_{DD}}{291.9\times10^{-6}} \frac{(V_{DD}+0.26)}{\times 0.96 (V_{DD}-0.7)^2}$$

$$t_{delay} = \frac{t_{PLH} + t_{PHL}}{2} \simeq \frac{V_{DD}}{(V_{DD}-0.7)^2} (0.00674 V_{DD} + 0.001908)$$

Energy delay product $= \frac{1}{2} C_{total} V_{DD}^2 t_{delay}$

$= 1.6\times10^{-15} V_{DD}^2 \cdot t_{delay}$

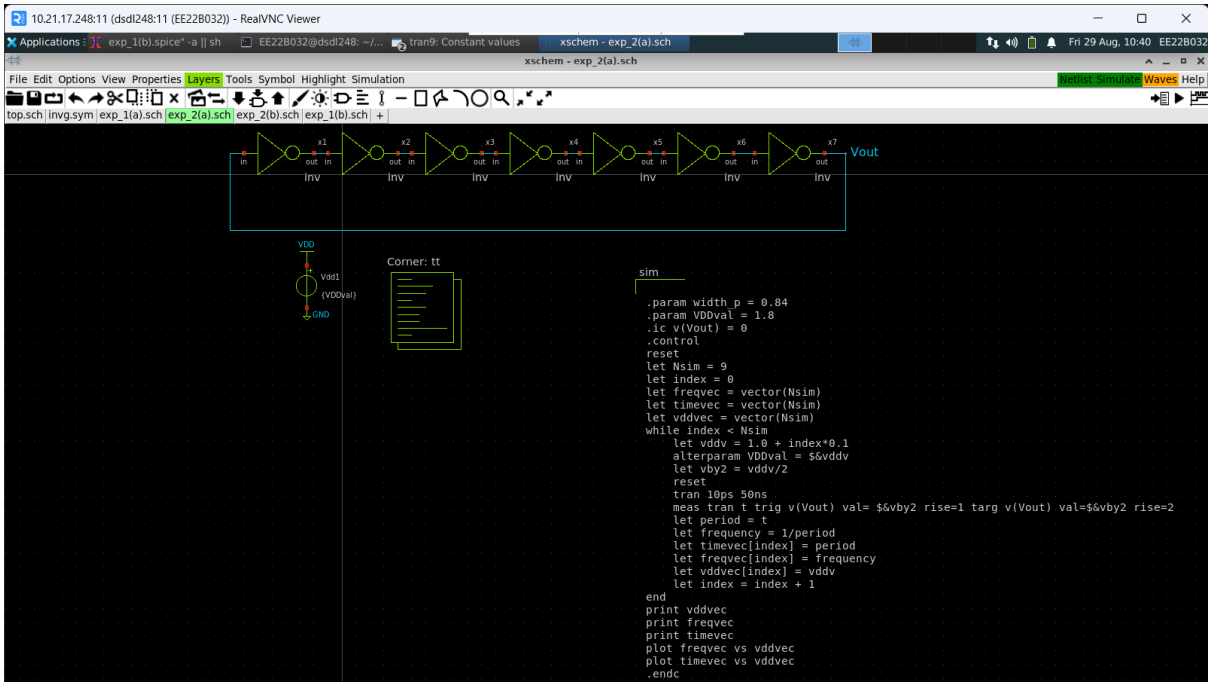
Analytical Estimates


Estimates
VDD tdelay Energy-delay product
1 - 96.08ps 153.74 x10^-27
1.1 - 64.08ps 124.07 x10^-27
1.2 - 47.96ps 110.54 x10^-27
1.3 - 38.53ps 104.18 x10^-27
1.4 - 32.41ps 101.64 x10^-27
1.5 - 28.16ps 102.4 x10^-27
1.6 - 25.07ps 102.68 x10^-27
1.7 - 22.72ps 105.06 x10^-27
1.8 - 20.88ps 108.27 x10^-27
* From above observed estimates, optimum VDD
≃ 1.4V //.

Optimum VDD

Question 2

7 inverters



Schematic

Index	vddvec
0	1.000000e+00
1	1.100000e+00
2	1.200000e+00
3	1.300000e+00
4	1.400000e+00
5	1.500000e+00
6	1.600000e+00
7	1.700000e+00
8	1.800000e+00

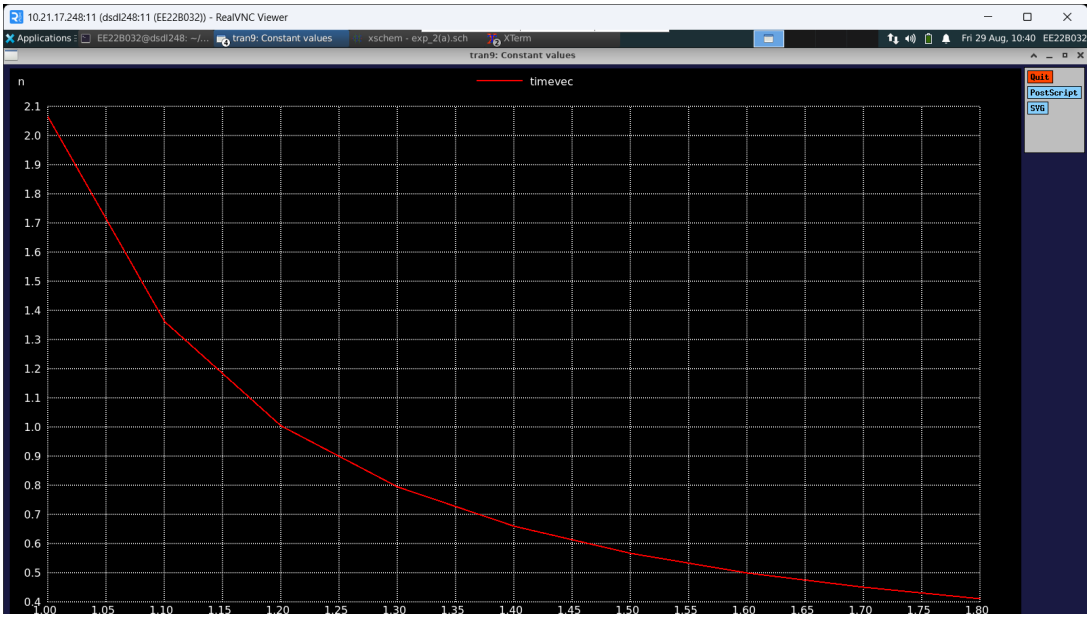
Constant values
constants Wed Jan 22 06:51:22 UTC 2025

Index	freqvec
0	4.843760e+08
1	7.338135e+08
2	9.951397e+08
3	1.257452e+09
4	1.513946e+09
5	1.761136e+09
6	1.996863e+09
7	2.217639e+09
8	2.423843e+09

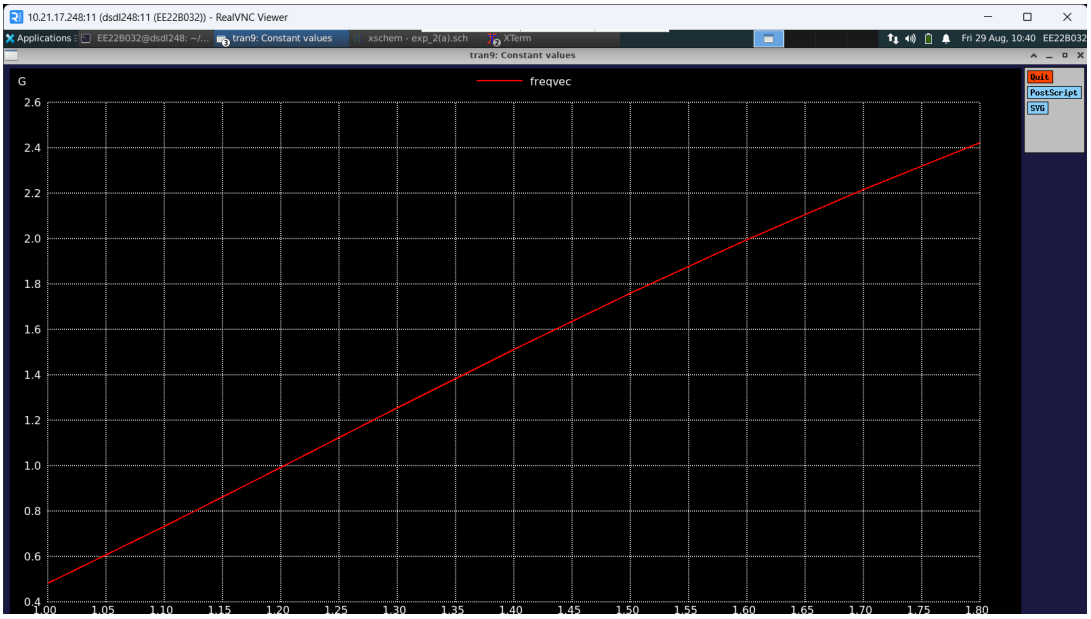
Constant values
constants Wed Jan 22 06:51:22 UTC 2025

Index	timevec
0	2.064512e-09
1	1.362744e-09
2	1.004884e-09
3	7.952588e-10
4	6.605257e-10
5	5.678152e-10
6	5.007856e-10

Measured Oscillating Frequency

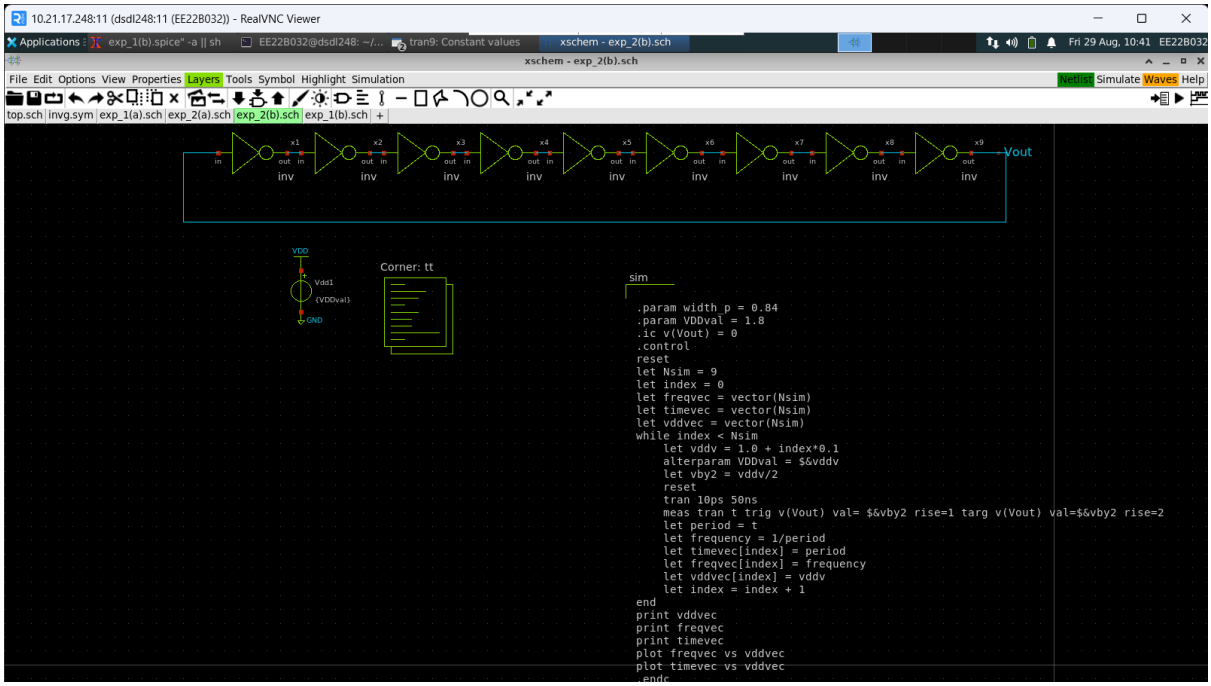


Time



Frequency

9 inverters



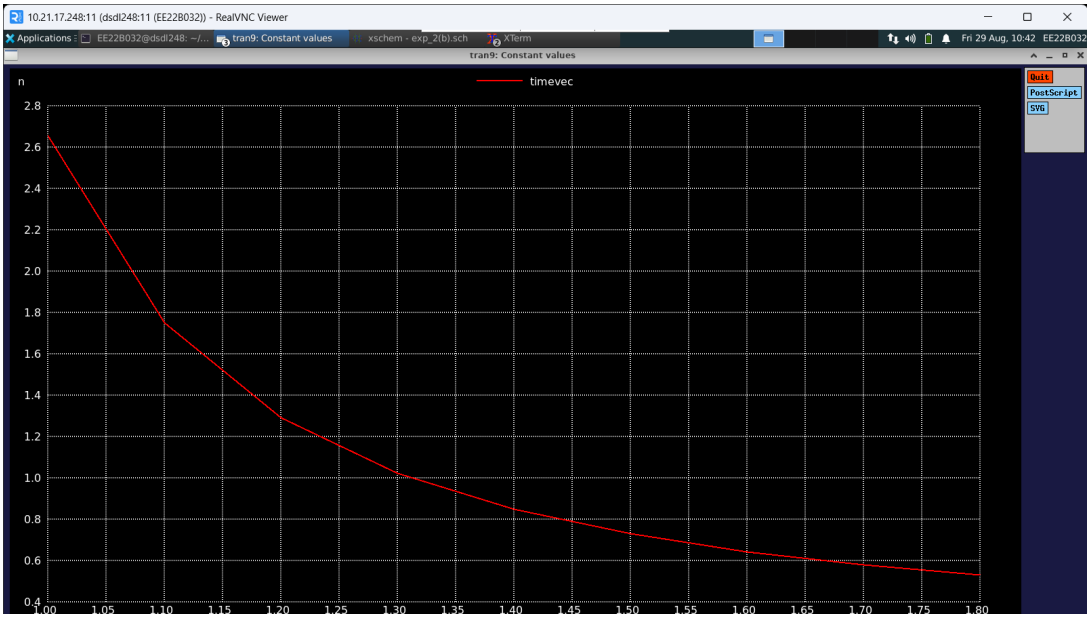
Schematic

```
--------------------------------------------------------------------------------
Index   vddvec
--------------------------------------------------------------------------------
0       1.000000e+00
1       1.100000e+00
2       1.200000e+00
3       1.300000e+00
4       1.400000e+00
5       1.500000e+00
6       1.600000e+00
7       1.700000e+00
8       1.800000e+00

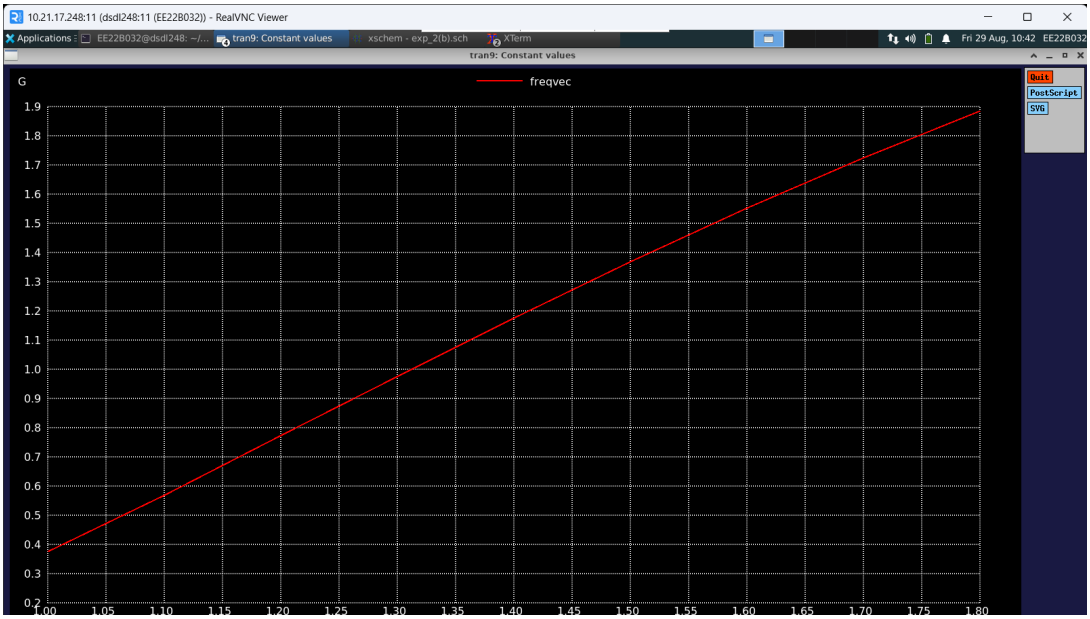
                            Constant values
                            constants  Wed Jan 22 06:51:22 UTC 2025
--------------------------------------------------------------------------------
Index   freqvec
--------------------------------------------------------------------------------
0       3.767027e+08
1       5.706903e+08
2       7.739087e+08
3       9.779033e+08
4       1.177547e+09
5       1.369719e+09
6       1.553635e+09
7       1.724698e+09
8       1.885375e+09

                            Constant values
                            constants  Wed Jan 22 06:51:22 UTC 2025
--------------------------------------------------------------------------------
Index   timevec
--------------------------------------------------------------------------------
0       2.654613e-09
1       1.752264e-09
2       1.292142e-09
3       1.022596e-09
4       8.492230e-10
5       7.300770e-10
6       6.436519e-10
```

Measured Oscillating Frequency



Time



Frequency